

June 2015
1506-1.08-1363

1.0 Introduction

In the newly modified master track link, the torque application requires a high degree of accuracy and also the required degrees ($90^{\circ} \pm 10^{\circ}$) are not easy to mark in the hexagonal bolt, generating an incorrect torque application to the track link. See Fig. 1.

This Best Practice provides a quick and safe solution for the precise marking regarding rotation degrees when applying the final torque on the tractor master track link bolts.

2.0 Best Practice Description

The current degree marking is done through the technicians' knowledge or expertise which generates errors that ultimately result in redoes for torque loss of or a failure in the bolts due to torque excess.

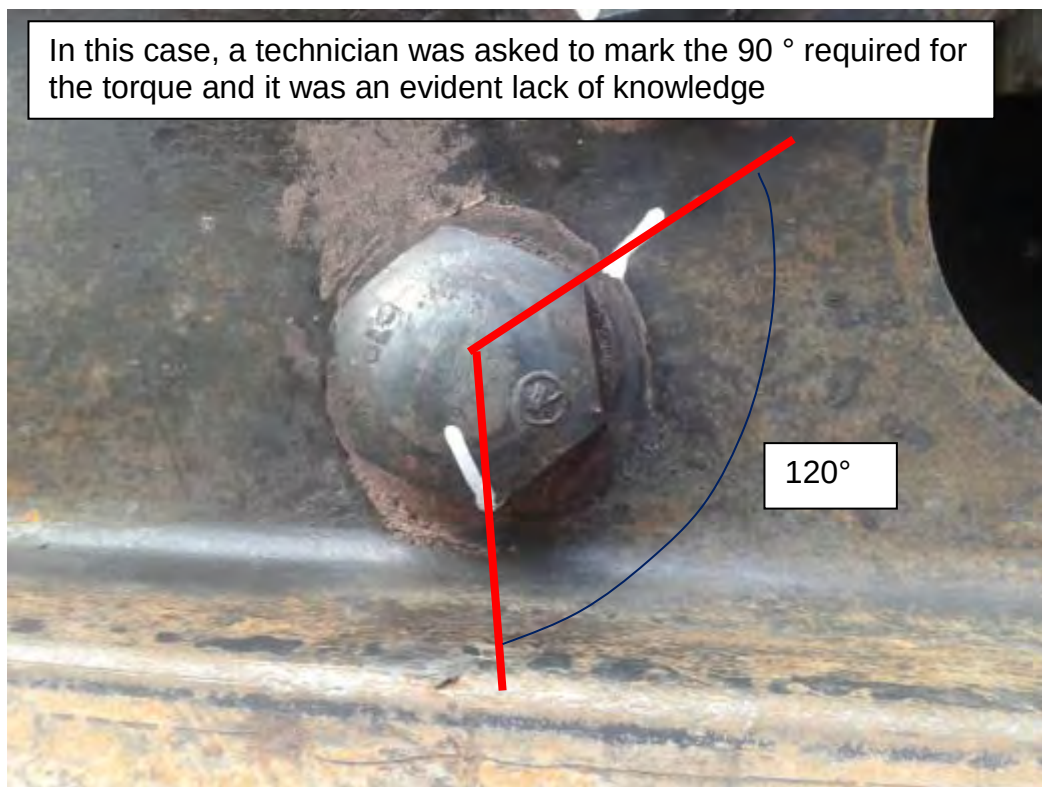


Fig. 1. Incorrect angle marking for the torque of the master track link.

To control the correct degree marking, a magnetic gauge was designed and manufactured that allows a better degree accuracy. Fig. 2

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Fig. 2. Gauge for degrees marking

Written on the visible surface of the gauge is the bolt specified diameter, the initial reference point of rotation and the difference, plus the most used degrees of torque for this.

This applies for the degree marking in the different models of bolts thru the different equipment and due to the fact that they are magnetic, they offer the convenience of placing them in any position.

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3.0 Implementation Steps

Logistics:

- Data collection regarding the procedure of degrees marking.
- Magnetic gauge kit for the degrees marking.
- Creation of a standard operating index card for controlling the torque.
- Disclose the Best Practice to the personnel.

Operation:

- Clean the area around the bolt.
- Apply the pre-torque to the bolts according to the suggested sequence.
- Install the gauge matching the initial reference point with one of the vertices of the hexagonal bolt head and mark this point. Also, mark the final point of rotation as indicated by the gauge (see Fig. 2).
- Remove the gauge and implement further rotation in the route from the starting point to the end point marked.
- Perform the procedure for all the bolts in the same order of the pre-torque application (see Fig. 3).

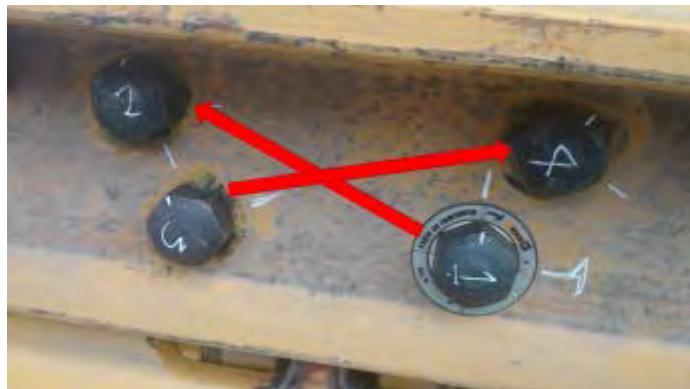


Fig. 3. Pre-torque and grades application sequence

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4.0 Benefits

- Reduction of 90% of redoes for torque loss of the master track link bolts. Under 65% of failures in less than 300 working days to 5.4% failure in the fleet of 37 machines. (See report attached).

Before

Failure percentage by work days

0>100 = 24,3%

100>200 = 27,%

200>300 = 10,8%

After

Failure percentage by work days

0>100 = 0% ▼

100>200 = 2,7% ▼

200>300 = 2,7% ▼

- More accuracy regarding degrees marking.
- Easy to use and it applies for any bolts that require further rotation
- The lack of dependence on the skill and / or knowledge of the technician to determine grades.
- Increased equipment availability.

5.0 Resources Required

- Magnetic gauges according to the drawings

6.0 Supporting Attachments / References

- Standard operation index card for the Master track link torque (D11, D10, D9, D6)

7.0 Related Best Practices

None

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8.0 Acknowledgements

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